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ORIGINAL ARTICLE

Demand estimation and market definition for broadband Internet services

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Abstract This paper analyses residential demand for Internet access in Austria with a focus on broadband Internet connections. Austria has cable network coverage of about 50% and is, therefore, a good candidate to analyse the elasticity of demand for DSL where cable is available and where it is not. We also include mobile broadband via UMTS or HSDPA in our analysis and estimate various nested logit models to derive conclusions for market definition. The estimation results suggest that demand for DSL is elastic and that cable networks are likely to be in the same market as DSL connections both at the retail and at the wholesale level. We discuss possible implications for the regulation of wholesale broadband access markets.

Keywords Estimation of discrete choice models · Broadband Internet access · Market definition · Regulation

JEL Classifications L51 · L96

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1 Introduction

Broadband Internet services are an integral part of the telecommunication industry—an industry that is subject to sector specific regulation. Whether there is a need for regulation of broadband markets at all and—related to this—how to properly define this market has been intensely debated in the US as well as in the EU.[1] One of the main issues regarded the question whether broadband Internet delivered via other platforms, in particular (upgraded) cable TV networks, is part of the same market as broadband Internet delivered via copper twisted pairs by means of DSL (digital subscriber line) technology.

This paper employs discrete choice models to estimate own and cross price elasticities of demand for DSL, cable, mobile broadband via UMTS/HSDPA and narrowband Internet access services in Austria. The used data are from a consumer survey commissioned by RTR (the Austrian Regulatory Authority for Broadcasting and Telecommunications) that was conducted in November 2006. There are around 2,800 observations that can be divided into two subgroups referring to consumers in two geographic areas. The division is based on information about cable and mobile network coverage. In one area all four Internet access types are available (DSL, cable, mobile broadband and narrowband). In the other area only DSL and narrowband are available. For each area, we estimate a nested logit model and derive own and cross price elasticities of demand.

We find that in the area where all Internet access types are available the demand for DSL, cable and mobile broadband is elastic with own price elasticities between −2.61 and −2.48. Demand for narrowband is also elastic (−1.93). In the area where only DSL and narrowband is available, demand for DSL is inelastic (−0.97) and so is demand for narrowband (−0.77). We interpret these findings such that the different broadband technologies constrain one another, while the constraint from narrowband on DSL is limited. A hypothetical monopolist test (or SSNIP-test)[2] shows that DSL and cable are likely to be part of the same market at the retail as well as at the wholesale level. We argue that this may necessitate a geographically differentiated regulatory approach to broadband markets.

The rest of the paper is structured as follows: Sect. 2 reviews the European discussion about market definition for broadband markets and the related literature. Section 3 gives an overview of the Austrian retail and wholesale market for broadband access including the current status of regulation. Section 4 describes the data and discusses the question how to allocate non-chosen alternatives to households. Section 5 describes empirical models of consumer behaviour, the estimation method and the model selection. Section 6 presents the estimation results for the two areas. Section 7 discusses the implications for market definition. Section 8 summarizes and discusses policy implications.

[1] For the US see, e.g. Crandall et al. (2002), for the EU see, e.g. European Commission (2004), Schwarz (2007), Inderst and Valletti (2007) and the discussion in Sect. 2.

[2] SSNIP it the acronym for small but significant increase in prices.

2 Motivation

The 2003 regulatory framework for electronic communications networks and services[3] of the European Union requires the national regulatory authorities (NRAs) of individual countries to periodically review a number of electronic communications markets which potentially may be subject to ex ante regulation. One of these markets is the market for wholesale broadband access.[4] Wholesale broadband access, also called 'bitstream access' (in particular if realised over a copper network), is a wholesale product which allows alternative operators to offer broadband Internet access to the final consumer without having an own access line. The alternative operator receives traffic at a higher network level (e.g. an ATM node) and will forward this traffic to the public Internet. It usually manages the customer relation, provides the Internet connectivity, e-mail addresses and web-space, and can influence the quality of service (e.g. by setting the overbooking factor).[5]

According to the 2003 regulatory framework, NRAs are required to periodically analyse the state of competition on the wholesale broadband access market. If an undertaking is found to have significant market power (SMP),[6] NRAs have to impose appropriate ex ante remedies[7] to prevent anti-competitive or exploitative abuses. Before the process starts, however, a relevant market has to be defined. The *Recommendation on Relevant Markets* quoted above is a starting point, however, NRAs have to check which products (and geographic areas) exactly to include or whether their national circumstances are such that they have to deviate from the Recommendation.

The instrument applied to define markets is, as in general competition law, the hypothetical monopolist test. This test asks whether, starting from the competitive level, a non-transitory 5–10% price increase would be profitable for a hypothetical monopolist in the market under consideration. The smallest set of products for which the price increase can be sustained constitutes the relevant market.[8]

Market definition of wholesale broadband access markets has attracted some attention since Oftel (now Ofcom) and Comreg (the NRAs of the UK and Ireland) have notified their decisions to the European Commission in 2003 and 2004.[9] One of the

[3] See Directives 2002/19/EC, 2002/20/EC, 2002/21/EC and 2002/22/EC, OJ L108, 24.4.2002.

[4] See Commission Recommendation of 11 February 2003 on relevant product and service markets within the electronic communications sector susceptible to ex ante regulation in accordance with Directive 2002/21/EC of the European Parliament and of the Council on a common regulatory framework for electronic communication networks and services, OJ L 114/45. ('Recommendation on Relevant Markets').

[5] For details on bitstreaming see ERG (2005).

[6] The concept of SMP is based on the concept of dominance in general competition law (see European Commissions Guidelines on market analysis and the assessment of significant market power under the Community regulatory framework for electronic communications networks and services ('SMP-Guidelines'—Official Journal 2002/C 165/03)).

[7] Ex ante remedies available to NRAs are listed in the access directive (Directive 2002/19/EC) and include obligations of access, non-discrimination, price control, accounting separation, and transparency.

[8] For a description of the HM-Test see, for example, Bishop and Walker (1999); OFT (2001), and §§ 49 et sqq. of the SMP-Guidelines.

[9] See Oftel (2003) and Comreg (2004). Decision on market definition and market analysis have to be notified to the European Commission which has a veto power.

main questions was whether access via cable networks (CATV-networks) forms part of the same market as access via copper networks (digital subscriber line—DSL). Whereas DSL wholesale products (bitstream acess) provided by the incumbent telecommunications operator (in most cases due to regulatory obligations or regulatory pressure) are available in many EU Member States (ERG 2005), wholesale broadband access via cable networks is only provided rarely. Therefore, a direct competitive constraint from cable on DSL at the wholesale level is unlikely to exist.[10]

However, as Ofcom, Comreg and later a number of other NRAs (although not all of them) argued, there is an "indirect constraint" from cable on DSL via the retail level. The argument is that a hypothetical monopolist for DSL access could not increase his bitstream prices profitably by 5–10% as this would also increase retail prices which would make customers switch from DSL to cable access at the retail level. This would also reduce access demand and if retail substitution is strong enough, the price increase would not be profitable. The elasticity of retail demand is therefore crucial not only for the definition of retail markets but also for the definition of the wholesale broadband access market. The European Commission, on the other hand, argued that such "indirect constraints" should not be considered at the stage of market definition, but only on the stage of market analysis (see European Commission 2004).[11] In addition, NRAs should provide evidence on retail demand elasticities to substantiate the existence of indirect constraints (see European Commission 2005).

While a number of papers have analysed the demand for Internet services or broadband services in particular,[12] only few papers have estimated the extent of retail demand elasticities for particular broadband Internet access types such as DSL and cable so far. Rappoport et al. (2003) use a nested logit discrete choice model to describe the demand for Internet access of residential customers in the US. They conclude that demand for DSL is elastic (own price elasticity of -1.462) and that therefore DSL and cable belong to the same retail market. Crandall et al. (2002) confirm these results (DSL own price elasticity of -1.184). Ida and Kuroda (2006) estimate a similar model for Japan including fibre (FTTH)—a rapidly growing access technology in Japan—in their choice set. They conclude that demand for DSL (at this time the main access technology with a share of 75%) is inelastic (own price elasticity of -0.846) but demand for cable and FTTH is elastic (own price elasticities of -3.150 and -2.500). They also find that the upper and lower end of the DSL market (very high and low bandwidths) are highly elastic as they directly compete with FTTH and cable on the high end and dial-up and ISDN (narrowband) on the low end. Finally, Pereira and Ribeiro (2006) estimate demand elasticities for broadband access to the Internet in Portugal, where the incumbent operator offers broadband access to the Internet both via DSL and cable modem. The authors' main aim is to analyze the welfare implications of the structural separation of these two businesses. The results suggest that households are very

[10] See however the notification of the Maltesian NRA, MCA (2006) which argues that there would be sufficient wholesale substitution if cable networks offered a wholesale product. This notification has been withdrawn later on, however.

[11] For a discussion on indirect constraints see also Inderst and Valletti (2007) and Schwarz (2007).

[12] See, for example, Madden and Simpson (1997), Varian (2000), Savage and Waldman (2004, 2005), Goolsbee (2006) and Goel et al. (2006).

sensitive to price variations in Internet access services. More specifically, the demand for broadband access is more elastic than the demand for narrowband access, with an estimate of −2.836 and −1.156, respectively. They conclude that broadband and narrowband access are substitutes. However, the demand for broadband access is less sensitive to the price of narrowband access than the demand for narrowband access to the price of broadband access, with cross price elasticities being 0.503 and 0.876, respectively. Considering DSL and cable individually yields even higher elasticities of −3.196 and −3.130 respectively.

Summing up, this evidence suggests that different types of broadband access constrain each other, however, to varying degrees.

3 The Austrian market for broadband Internet services

By the end of 2006, 52% of all private households in Austria had an Internet connection. While narrowband connections (dial-up and ISDN) still have a significant share (19% of all households in 2006), the share of broadband connections is increasing rapidly, from 10% in 2002 to 33% in 2006.[13] The Austrian market for broadband Internet services therefore is—like in many other countries—characterised by high and steady growth.

Broadband Internet via cable networks became available in 1996 and DSL followed in 1999. By the end of 2006, there were 1.36 Mn. broadband connections, more than 1 Mn. of which were held by residential customers. The broadband penetration in Austria was 39% (households and businesses, RTR 2007) and slightly above the EU average.

The cable network coverage is approximately 50% of all households and relatively high compared to most other European countries.[14] There are more than 100 cable network operators which offer broadband services in different regions of Austria (cable networks usually do not overlap). However, almost 90% of all cable connections are offered by six bigger operators. The DSL coverage is, like in most other EU countries, above 90% of all households.

In September 2006, the shares of the different infrastructures on the market for fixed broadband connections were as follows: DSL: 61%, cable: 37%, other (fixed wireless access, fibre): 2% (RTR 2007). While most business users prefer DSL over cable, cable holds a strong position in the residential market. Most DSL and cable operators offer a menu of three to five (and sometimes more) tariffs, which vary by price, download speed and download volume.

Since 2003, mobile broadband via UMTS and since 2006, mobile broadband via HSDPA is available. By the end of 2006, there were about 220,000 mobile broadband connections, more than half of which were used by business customers—often complementary to fixed access. Residential customers on the other hand seem to use mobile broadband rather as a substitute than a complement. Mobile broadband via HSDPA is usually available in cities with more than 5,000 inhabitants, where in most

[13] See Statistik Austria (2006).

[14] Exceptions are the Netherlands, Belgium, Luxembourg and Switzerland with almost full cable coverage.

cases also cable networks exist. Hereby, Austria is one of the leading countries in the deployment of mobile broadband services via UMTS/HSDPA.

Mobile tariffs are designed somewhat differently from fixed network tariffs insofar as price does not vary with download speed but only with download volume. While on-off connection fees were usually lower than in the fixed network by the end of 2006, volume was still more expensive.

There are two regulated wholesale products based on which alternative operators, which do not own infrastructure all the way to the final consumer, can enter the retail market: local loop unbundling (LLU) and bitstream access (described in Sect. 1), both of which allow alternative operators to offer DSL connections on the retail market. While more own infrastructure is needed for local loop unbundling, there is also more value added and there are more degrees of freedom for designing the products (e.g. bundles with voice telephony, etc.). Due to this wholesale regulation and platform competition in several areas, the retail market is considered competitive by the Austrian NRA.

4 Data and descriptive statistics

The data we use is from a survey commissioned by RTR (the Austrian National Regulatory Authority) which was conducted in November 2006. 4,029 households were interviewed about the type and characteristics of the Internet connection they use as well as their monthly expenses. Individual specific data such as age, education and household size were also collected. After eliminating missing and implausible values we are left with 2,825 observations. For the estimation, these observations are divided into two sub-samples: One for the area where all four Internet access technologies (DSL, cable, mobile, and narrowband) are available and another one for the area where only DSL and narrowband are available.[15] The number of observations is reported in Table 1. The share of the different types of access in the sample is unbalanced. A comparison of the data with other surveys (Statistik Austria 2006; Integral 2006) and data from (RTR 2007) shows that there are too many DSL households and too few narrowband and cable households. We therefore use weights in order to correct for that. We weigh the observation such that the weighted distribution of access types corresponds to the one from a micro census carried out by the central bureau of statistics (Statistik Austria 2006), which is also consistent with the data in Integral (2006) and RTR (2007).

Table 2 reports the descriptive statistics of the main product-specific variables which we use in the analysis.

As can be seen, users on average spend less on DSL than on cable products, however, the DSL products come—on average—with lower speed and volume. Cable products are also much more frequently bought with flat rate (57.80%) than DSL products (8.40%). For mobile products it is difficult to determine a download rate as

[15] Data on cable network and DSL coverage are available from the operators. Data on mobile coverage are not available. However, it can be concluded from the operators' press releases that HSDPA coverage by end of 2006 was available in urban areas where usually also cable networks are present. We therefore use the cable network coverage as a proxy for the mobile broadband coverage.

Table 1 Number of observations

	DSL	Cable	Mobile	Narrowb.	No internet	Total
Total unweigted	644	234	29	236	1,682	2,825
Total weighted	375	278	34	452	1,650	2,789
Cable/mobile area unweighted	387	234	29	139	964	1,753
Cable/mobile area weighted	229	278	34	269	968	1,778
DSL/narrowband area unweighted	257	0	0	97	718	1,072
DSL/narrowband area weighted	146	0	0	183	682	1,011

Table 2 Descriptive statistics—product specific variables

	Mean	Std. dev.	Min	Max
Price in € per month				
DSL	31.73	9.40	9.90	73.00
Cable	40.73	13.87	19.00	75.00
Mobile	32.20	10.24	9.50	59.00
Narrowband	18.98	12.94	4.00	60.00
Download speed in kbit/S				
DSL	1,365	999	210	6,144
Cable	3,180	2,492	128	16,384
Mobile	900	0	900	900
Narrowband	56	0	56	56
Download volume included in MB (for non-flat rate products)				
DSL	1,561	2,418	250	21,000
Cable	4,187	5,494	400	20,000
Mobile	777	844	250	4,000
Narrowband	0	0	0	0
Share of flat rate products (dummy_flat)				
DSL	8.40%			
Cable	57.80%			
Mobile	0.00%			
Narrowband	0.00%			

Table 3 Individual specific variables

	DSL	Cable	Mobile	Narrowb.	No internet
Age (head of household—mean)	44.90	42.67	40.97	45.54	60.50
Household size (mean)	3.1	2.7	2.7	3.1	2.0
Education: compulsory school	37%	28%	40%	30%	70%
Education: high school without graduation	22%	14%	21%	25%	17%
Education: high school with graduation	25%	33%	26%	29%	10%
Education: university degree	15%	25%	12%	16%	3%
Gender: female	45%	42%	43%	54%	68%

it depends, for example, on the location and the number of users in the cell. We have taken 900 kbit/s as a maximum value a consumer could expect to get by end of 2006. The included volume for mobile broadband is much lower than for fixed broadband connections. Individual specific variables are reported in Table 3.

The discrete choice approach we use for the analysis of demand requires us to allocate each household all four Internet access types, which immediately raises the question of how to determine the price and characteristics of the non-chosen alternatives. This is a crucial point as it can significantly influence the result of the analysis. We describe our approach by type of Internet access.

4.1 Narrowband

Narrowband prices are most difficult to match as it is quite hard to say how much a broadband or 'no Internet' household would spend on narrowband services which are metered by minute. Narrowband expenses are also hard to explain by the individual specific variables available, however, they seem to vary with region and age. We therefore form eight groups (four regions combined with age smaller or larger than 50), calculate group averages and impose those on broadband and 'no Internet' households.[16] Speed and included download volume does not vary for narrowband households (the maximum speed is always 56 kbit/s and the volume included is always zero).

4.2 Cable

We use the tariffs from the web pages of seven big cable operators which cover all of the nine federal states of Austria.[17] We assume that smaller cable operators (which have very low market shares in total) charge prices similar to the bigger local operator we use. All operators offer different products varying by bandwidth and included volume for low, medium and high usage. 'No Internet' households and narrowband households who spend less than €25 per month are assigned the 'low user' package of the local operator. Narrowband households which spend more than €25 per month are assigned the 'medium' package. DSL households are assigned the package which is closest in bandwidth and mobile households are assigned the package closest in price.

4.3 DSL

DSL is basically done in the same way as cable. We use the tariffs from the web pages of the largest three DSL operators.[18] Geographic differences result from two operators which offer DSL access based on local loop unbundling in different parts of Austria. Where only Telekom Austria—the incumbent DSL operator—is present, we take the packages of Telekom Austria. Where also one or two of the LLU operators are present, we take average values (weighted by national market shares).

[16] This might cause an endogeneity problem. However there does not seem to be a good alternative and since we use this method only for narrowband we think that this should not be too problematic. This procedure is also applied by Ida and Kuroda (2006).

[17] These operators are: UPC, LIWEST, Salzburg AG, kabelsignal, B.net, Telesystem Tirol and Cablecom.

[18] These operators are Telekom Austria, UPC/Inode and Tele2.

Table 4 Results of testing the null hypothesis 'H_0: difference between actual paid amounts and monthly fixed charges for DSL and cable = 0'

Volume	Obs.	Mean	Std. Err.	t-value
<500	268	4.003	0.565	7.083***
[500, 1,000]	233	4.903	0.693	7.071***
[1,000, 5,000]	100	1.910	1.031	1.853*
>5,000	184	−0.314	0.997	−0.315
Flat rate	152	−0.414	1.094	−0.378

***, ** and * denote significance on 1%, 5% and 10% level

For both DSL and cable, there is a problem with households which are assigned a non-flat rate tariff. It can be assumed that these households—on average—have to pay an additional amount per month for exceeding their included download volume. We solve this problem by comparing actual amounts paid to monthly fixed charges for DSL and cable households over several groups of included download volume. We find that the mean difference between actual amounts paid and monthly fixed charges is significantly different from zero for low download volumes. The result of the t-test is depicted in Table 4. We use the mean as 'mark-up' for matched DSL and cable products if it is significantly different from zero at least at the 10% level.

4.4 Mobile

Mobile products are assigned according to the monthly charge which is currently paid by the household. We use averages of the prices of the four mobile operators weighted by market share. The download speed does not vary for mobile broadband (the maximum speed is assumed to be always 900 kbit/s).

5 Empirical strategy

This section discusses the empirical strategy to obtain estimates for the elasticity of demand with the help of which we are going to define the market for broadband Internet. We describe the empirical models of consumer behaviour and discuss the assumptions necessary for identification, the estimation method and model selection.

5.1 Empirical models of consumer behaviour

To empirically analyse consumer behaviour, we assume a random utility model of Internet access in which consumers choose from a set of five choices. These are no Internet access, dialup Internet access (narrowband), cable network, DSL and mobile Internet access. In some (predominantly rural) areas cable network and mobile broadband are not available. Here we model consumers to choose from a set of three choices (no access, narrowband, DSL). The utility a consumer derives from a particular product depends on characteristics of the consumer and on the characteristics of the product.

To account for characteristics that are unobserved by the econometrician, the utility of consumer i for product j is of the form

$$U_{ij} = V_{ij} + \varepsilon_{ij}, \tag{1}$$

where i and j are the indices for consumer $i, i = 1, \ldots I$, and product j, $j = 1, \ldots J$, and where the term V_{ij} reflects the deterministic part of consumers' utility. The error ε_{ij} is a residual that captures for example, the effects of unmeasured variables or personal idiosyncrasies. It is assumed to follow an extreme value distribution of type I.

Consumers are assumed to purchase that Internet access that gives them the highest utility. The probability P_{ij} that consumer i purchases product j is equal to the probability that $U_{ij}$is larger than the utility consumer i experiences from any other product, i.e. $U_{ij} > U_{ij'}$ for all $j' \neq j$. This probability is equal to

$$P_{ij} = P[U_{ij} > U_{ij'} \forall j' \neq j] = P[\varepsilon_{ij'} - \varepsilon_{ij} \leq V_{ij} - V_{ij'} \quad \forall j' \neq j] \tag{2}$$

Under the assumption that ε_{ij}follows an extreme value distribution of type I, the probability P_{ij}has a closed form solution (McFadden 1974). It is equal to

$$P_{ij} = \frac{e^{V_{ij}}}{\sum_{j'} e^{V_{ij'}}}. \tag{3}$$

This is the well-known conditional logit model. Within this model, we have to assume the independence of irrelevant alternatives (IIA). To additionally model correlations between choices, nested logit models have been developed (see, for example, Maddala 1983; Train 2002 or Greene 2003). The nested structure of the Internet choice places less structure on the decision process than the conditional logit model. The IIA assumption across all alternatives can be dropped, but it still has to hold between alternatives in one nest. Choices in one nest are assumed to be similar in unobserved factors. With nested logit models one usually obtains more reasonable substitution patterns and estimated cross price elasticities. We use two- and three-layer specifications to model the choice of Internet access. Figure 1 depicts a three layer model for the area where all types of Internet access are available and a two layer model for the area where only DSL and narrowband are available. In both models, the outside option is 'no Internet'. For the three layer model we report choice probabilities and elasticities in Appendix B.

We specify the deterministic part V_{ij} as a linear function of consumer characteristics, Z, and product characteristics, X including the price of the product, such that

$$V_{ij} = \gamma Z_i + \beta X_j, \tag{4}$$

where γ and β are the parameters to be estimated. We assume a simple specification so that the β's are constant over all choices j. Consumer characteristics are for example, age, educational dummy variables or household size. Product characteristics are the price, the download rate and the download volume. Among the various characteristics

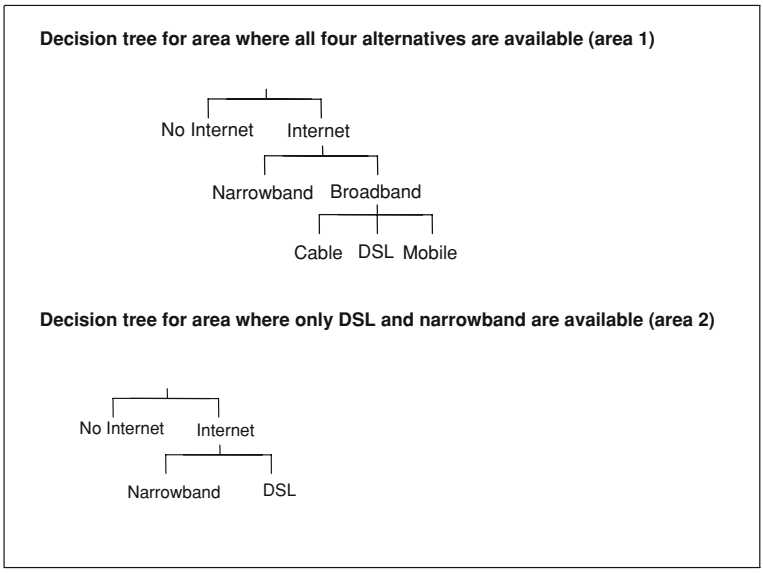

Fig. 1 Decision trees for the nested choice model

that enter the demand, the price is particular important as we obtain own and cross price elasticities from the estimated coefficient.

We also tested for the interaction of consumer and product characteristics, i.e. specifications so that the β's vary over choices j. We found that the linear specification is better with respect to model selection criteria, but also that in particular the estimated demand elasticity of DSL do not change very much. We therefore present only the results from the simple specification given in (4). This specification is also in line with Rappoport et al. (2003) and Pereira and Ribeiro (2006).

5.2 Identification

To be able to identify the parameters γ and β, sufficient variation in prices and in the characteristics of consumers, who experience similar prices, is necessary. As there are only a finite number of tariffs available for consumers, we obviously observe only a finite number of prices in our sample. If the number of tariffs were too small, we would not be able to identify the estimated coefficient on prices. Fortunately, this is not the case; the smallest standard deviation for the four different access types is close to 10 and there is a wide range of prices, the smallest range (difference between minimum and maximum) of €49.5 observed for the mobile category. In addition to this, our data show that there are 113 different prices. With the choice narrowband, we observe 55 different prices. The respective numbers for cable, DSL and mobile are 37, 55 and 13. This provides us with enough variation to identify the coefficient on prices.

We also observe enough variation in the characteristics of consumers, who experience similar prices. Without this variation, it would not be possible to identify the

estimated coefficient on prices and consumer characteristics or any interaction of prices and consumer characteristics. For example, if consumers in the lowest income class always choose the lowest price and consumers in the highest income class always choose the highest price, income perfectly predicts prices. If there is additionally insufficient variation in prices, income would also perfectly predict the choice of Internet access. To assess the variation in the data, we divide our data on prices in ten different price ranges, calculate the variation of consumer characteristics in each price range and compare it to the variation in the whole sample. For each price range and each consumer characteristics like age, education and household size, we observe the same distribution of values as for the whole sample. Only for very high prices, the variation in consumer characteristics decreases. This provides us with enough variation to identify the coefficient on prices and the coefficient on consumer characteristics.

5.3 Estimation method and model selection

We estimate the above described models with sequential maximum likelihood. The estimation is decomposed into two or three stages depending on the model (two or three layer model) and each stage corresponds to a branch of the tress depicted in Fig. 1. Another possibility would be to directly maximize the likelihood function associated with the nested logit models of Fig. 1. However, the direct maximization (of the three layer model in particular) proofed to be infeasible to obtain convergence. We therefore estimated all models with sequential maximum likelihood.

We weight the observations as discussed in Sect. 2. For comparison, we also estimated the models with the unweighted sample and obtained similar results for the estimated coefficients. In particular, we obtained nearly identical estimated values for the elasticities of demand.

We also estimated different tree structures and selected the best model based on Hausman specification tests. The alternative nesting structures we tested for are depicted in Fig. 2. The first nesting structure assumes cable, DSL and mobile to be in one nest and no Internet, narrowband and broadband to belong to the first layer. The second nesting structure assumes narrowband, cable, DSL and mobile to be to one nest and no Internet and Internet to belong to the first layer. Because the alternative specifications incorporate mutually exclusive dimensions of product differentiation,

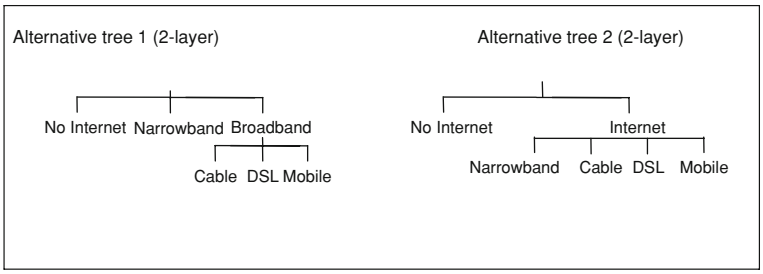

Fig. 2 Alternative decision trees for the nested choice model

Hausman tests are appropriate. Goldberg (1995), for example, applies the same testing procedure. If this was not the case, we would have to proceed according to Bresnahan et al. (1997).

We tested for the IIA property in the first layer of alternative tree 1 and in the second layer of alternative tree 2. We rejected both models, i.e. no Internet, narrowband and broadband do not belong to one nest as well as narrowband, cable, DSL, and mobile do not belong to one nest. In addition to the alternative nested logit models depicted in Fig. 2, we estimated a conditional logit model with all alternatives (no Internet, narrowband, cable, DSL, mobile in area1 and no Internet, narrowband, DSL in area 2) in one nest and test for independence of irrelevant alternatives (IIA). For both areas, we rejected the conditional logit model.

6 Estimation results

This section presents the estimation results. We first describe the estimation results for the region where all four types of Internet access are available and then describe the results for the region where only DSL and narrowband are available.

6.1 Area 1: DSL, cable, mobile, narrowband

The estimation results of the nested logit model for area 1 are given in the Table 7a, b and c (see the Appendix A). Table 7a gives the results for the bottom stage, Table 7b and c those for the second and the first stage, respectively.

At the bottom level, the independent variables are price, the download rate, the download volume, a dummy variable for flat rate tariffs and two dummy variables indicating the fixed term of DSL and mobile alternatives. Furthermore, we use age, household size, educational dummies (with the highest education—university degree—as the omitted dummy variable), a gender dummy and a dummy for the capital city Vienna. Individual specific variables are interacted with dummies for DSL and mobile respectively.

The price has the expected negative sign and is highly significant. The download rate and the download volume both have the expected positive signs and they are significantly different from zero at conventional significance levels. The flat rate dummy is negative and significant which might seem odd but can be explained as follows: In order to be able to use the observation in the estimation, we have to assign some value to the variable 'volume' even if the product has a flat rate. We have chosen a value of 55,000 MB (55 GB), which is just a little larger than the largest volume of non-flat rate products. The negative sign on the flat rate dummy variable can now be interpreted such that the actual utility a consumer derives from a flat rate measured in volume is less than 55,000 MB.

Most individual specific variables are insignificant at the individual level and therefore do not appear to influence the choice between different broadband technologies. Exceptions are the dummy variables for Vienna, which indicate that it is less likely to have DSL or mobile compared to cable in Vienna. This can be explained by the strong position of the cable network operator UPC in Vienna. Many Viennese households

also have cable TV and obviously prefer to buy broadband from the same operator. When we test for the joint significance of all consumer characteristics, we obtain a value of 49.06 for the Chi-squared statistic indicating a joint significance at the 99% level. We therefore prefer this specification over a specification without consumer characteristics; a specification that gives nearly identical demand elasticities.[19]

The goodness of fit of the model is evaluated using the McFadden R^2 or the likelihood-ratio index, which compares the likelihood for the intercept only model to the likelihood for the model with the predictors. The value of the McFadden R^2 of the first stage is 0.34.[20]

At the second stage we use a dummy variable for narrowband and the same individual specific variables as in the bottom level (interacted with a dummy variable for narrowband). However, only the fixed-effects dummy for narrowband is individually significant at a better than the 10% level. When we again test for the joint significance of all consumer characteristics, we obtain a value of 13.02 for the Chi-squared statistic indicating a joint significance at the 5% level. The McFadden R^2 shows that the model performs moderately well with a value of 0.25.

At the third layer, we use a dummy variable for no Internet and again the same individual specific variables (interacted with a dummy variable for no Internet). All variables (with the exception of the Vienna-dummy) have the expected sign and are significant at the 1% level. The McFadden R^2 is relatively high (0.41).

Table 7b and c also report the estimated coefficients of the inclusive value. The estimated coefficient of the inclusive value of the third stage is 13.46 and significantly different from zero at the 1% level. The estimated coefficient of the inclusive value of the second stage is 1.57 and significantly different from zero at the 1% level.

Coming now to the own price elasticities of access demand, the three-layer model implies elasticities for broadband services in the range of −2.617 to −2.481 (see Table 9 in the Appendix A). The elasticity of DSL services is −2.545 indicating that a one percent increase in the price decreases the demand for DSL services by 2.5%. The corresponding figures for mobile and cable services are −2.481 and −2.617, respectively. The lowest elasticity is estimated for narrowband services, which is equal to −1.679.

The results indicate that demand for all services is elastic. However, broadband services appear to be more elastic than narrowband services. An interpretation of this may be that different broadband services (in particular DSL and cable) constrain each other, while those consumers still using narrowband do not consider broadband as an equally good substitute.

While the cross-price elasticities derived from our estimation results are of the right sign, they are of relatively small magnitude ranging from 0.183 to 0.402. However,

[19] Pereira and Ribeiro (2006) also observe statistically insignificant effects of demographic variables. They prefer to drop the individual variables and to use a mixture component of these factors. Since our tests reveal the joint significance of these factors, we keep them in the equation. An alternative strategy would be to analyze the significance of all conceivable combinations of the demographic variable. We do not choose this option due to its ad hoc nature.

[20] The McFadden R^2 is useful for comparing models. The model fit can also be based on measures of information such as Akaike's information criterion (AIC) and the Schwarz information criterion (BIC). The values of AIC and BIC are 0.383 and 0.397, respectively.

they do not constitute an unusual exception to the existing estimates from other studies. The cross-price elasticities reported by Crandall et al. are 0.591 and 0.415 for Cable Modem and DSL. Rappaport et al. report higher cross-price elasticities of 0.766, whereas the estimates are 0.503 and 0.135 for narrowband and broadband in Pereira and Ribeiro (2006). Our estimates are on the lower end of these estimates, however they still suggest dynamic interactions between the different access types.

6.2 Area 2: DSL and narrowband

As it was mentioned earlier, cable networks and mobile broadband are not available in all regions. Mobile broadband via HSDPA is only available in cities with more than 5,000 inhabitants where in most cases also cable networks exist. In this section we consider only those observations where these two alternatives are not in the choice set of consumers.

For this sample we estimate a two-layer nested logit model where the consumers decide first whether they would like to have Internet access or not. At the bottom level, the consumers are then faced with the choice between DSL and narrowband access.

At the bottom level (Table 8a in the Appendix A), the independent variables are price, the download rate, the download volume, a dummy variable for flat-rate tariffs and a dummy variables indicating the choices for narrowband. The individual specific variables age, household size, educational dummies and a gender dummy have been interacted with the narrowband dummy. The estimated coefficient on the price is significantly negative tough its magnitude is substantially lower than in the three-layer specifications reported for area 1. The coefficients on download rate and download volume are also significantly different from zero at conventional levels and have the expected sign. The McFadden R^2 is smaller than for area 1 (0.13).

On the second stage we again use a dummy variable for no Internet and the same individual specific variables as before (interacted with the no Internet dummy variable). Most variables are significant and have the expected sign. The fit of the model is relatively good (McFadden R^2 of 0.37). The estimated coefficient of the inclusive value is 1.91 and significantly different from zero.

Turning now to the elasticities implied by this sample (see Table 9 in the Appendix A), we note that the own price elasticity of demand for DSL is equal to -0.97 and -0.77 for narrowband. This is a rather large decline in the absolute value compared to the elasticities for area 1. This suggests that cable and mobile broadband significantly constrain DSL in area 1, while the constraint from narrowband on DSL is limited. Since also the narrowband elasticity declines, it appears that cable and mobile not only constrain DSL in area 1, but to some extent also narrowband.

Since area 2 consists of regions which are more likely to be rural areas, the sharp decline in elasticities is by and large consistent with our a-priori expectations. Survey results show that the main reason behind having Internet access is because of enjoyment and to save time and money (Savage and Waldman 2005). In rural areas, we expect Internet access to be more important due to long distances to cities where consumers have other options to achieve all these goals. Therefore, we expect inhabitants of rural areas which have an Internet connection to be less sensitive to higher prices so that the expected demand profile will be more inelastic.

7 Implications for market definition

The instrument used for market definition is the hypothetical monopolist test. This test asks whether, starting from the competitive level, a non-transitory 5–10% price increase would be profitable for a hypothetical monopolist in the market under consideration. In the case of wholesale broadband access markets, the question is, whether a hypothetical monopolist for DSL lines on the wholesale level could profitably increase prices by 5–10% above the competitive level. This can be implemented by comparing profits before and after the price increase. If profits decrease after a 5–10% price increase, i.e.

$$(w_1 - c)x_1 > (w_2 - c)x_2 \tag{5}$$

this means that the closest substitute to DSL has to be included in the market. In (5), w denotes the wholesale price, c constant marginal costs and x the number of DSL lines sold; subscripts 1 denote prices and quantities before and subscripts 2 prices and quantities after the price increase. Fixed costs are assumed to remain unchanged in the short to medium run and therefore do not have to be considered in the profit comparison.

Section 6 has derived an estimate of the retail demand elasticity for DSL services. However, in order to define the market for wholesale broadband access we need the elasticity of demand at the wholesale level. As demand for inputs at the wholesale level is derived from demand at the retail level, the elasticity of demand at the wholesale level will be related to the elasticity of demand at the retail level.[21]

Under the assumptions that (i) one unit of the wholesale input is used to produce one unit of the retail good, (ii) there is no alternative input at the wholesale level and (iii) wholesale and retail supply is competitive, the relation between retail and wholesale demand elasticity is

$$\varepsilon_W = \frac{w}{p}\varepsilon_R, \tag{6}$$

where ε_W is the wholesale elasticity ε_R the retail elasticity, w the wholesale price and p the retail price. Assumptions (i) and (ii) are uncritical in the case of wholesale broadband access markets: One bitstream-access line is used to provide one DSL line at the retail level, and other infrastructures do not offer wholesale access or only to a limited extent. Assumption (iii) also appears to be justified in the case of Austrian broadband markets. While in particular wholesale supply might not be competitive, there is ex ante regulation in place (on the markets for local loop unbundling and wholesale broadband access)[22] which should—together with some competitive pressure from

[21] For a discussion on the relation between wholesale and retail demand elasticities see, for example, Inderst and Valletti (2007) and Schwarz (2007).

[22] On the market for local loop unbundling, the incumbent operator is obliged to offer unbundled local loops to alternative operator and to set cost-oriented prices according to the LRAIC (long run average incremental costs) principle. On the market for wholesale broadband access, the incumbent operator is obliged to offer bitstream products while prices are regulated according to the retail minus principle.

cable and mobile broadband in some areas—prevent the incumbent operator from exercising his market power at the wholesale as well as at the retail level. The Austrian Regulatory Authority RTR found the retail market to be effectively competitive (given the above mentioned ex ante regulation). We therefore also consider the prevailing (retail and wholesale) prices as a good benchmark for the competitive prices. The use of prevailing prices as a starting point for the hypothetical monopolist test in regulated markets is also recommended by the European Commission (see European Commission 2002, § 42).

The wholesale elasticity therefore can simply be calculated by multiplying the retail elasticity with the share of wholesale costs in the retail price. This share can be estimated to be about 75% in the case of bitstream products in Austria based on a comparison of wholesale (bitstream) and retail prices of the incumbent operator in November 2006. If we take the retail elasticity from the three-layer model, −2.545, the wholesale elasticity would be −1.909.[23]

Estimates of marginal costs are based on detailed (but confidential) cost accounting data from the incumbent operator and several alternative operators available to RTR. Table 5 shows the cost categories for the provision of wholesale broadband access (bitsteraming) on a DSL network. The different categories are divided into three groups. Some costs vary per customer (e.g. modem and installation). These costs are considered as variable. Some costs do not vary by individual customer, but over a limited number of customers (e.g. the DSLAM[24] or some personnel costs). These costs are considered as "step fixed". Depending on how many customers the hypothetical monopolist loses as a consequence of a 5–10% price increase, they might be considered either as variable or as fixed (the classification of costs as step fixed or fixed may also depend on the time frame considered). To allow for this they enter in the calculation once as fixed costs and once as variable costs. Some costs are fixed in the short to medium run even if there is a big change in the number of customers. These costs include backhaul installation, equipment, capital costs and overheads. Considering step-fixed costs as fully fixed and fully variable, respectively, results in a range of 20–40% variable costs in total costs.[25]

With this information the change in profits after a 5–10% price increase can be calculated as in (5). The result of the comparison is depicted in Table 6.

It can therefore be concluded that a 5–10% price increase from the competitive level would not be profitable for a hypothetical monopolist of DSL lines at the wholesale level due to substitution at the retail level. This means that the next best substitute (at the retail level) would have to be included into the relevant market. Within the broadband nest, cable has the highest cross-price elasticity. This also appears plausible taking into account similarities between tariffs and product characteristics as well

[23] This estimate is based on demand of residential users. The demand of business users for DSL is likely to be more inelastic, however, business users make up only about 20% of all DSL lines and this share is decreasing.

[24] Digital Subscriber Line Access Modem. It is operated at the main distribution frame and can typically supply several hundred customers.

[25] This is a rather wide range. However, if the conclusions even hold for such a wide range (which is the case, as can be seen below), they can be considered as relatively robust.

Table 5 Cost categories for providing bitstream access

Cost category	Variable/step fixed/fixed
Access*	
Customer installation	Variable
DSLAM	Step fixed
User modem	Variable
Backhaul	
Installation	Fixed
Connection	Step fixed
Equipment	Fixed
Other	
Maintenance	Some fixed, some step fixed
Personnel	Some fixed, some step fixed
Capital costs	Fixed
Overheads	Fixed

* The access line itself dos not enter in this calculation as these costs are covered by the subscription fee for voice telephony (DSL broadband access is in more than 95% of all cases consumed together with voice telephony). If they would enter, this would increase the share of fixed costs and reinforce the conclusions below

Table 6 Comparison of profits before and after a 5–10% price increase

	20% marginal costs	40% marginal costs
$\Delta\pi$ after 5% price increase	−3.9%	−2.0%
$\Delta\pi$ after 10% price increase	−9.0%	−5.6%

as current penetration rates. It therefore appears that cable exerts the highest competitive constraint on DSL and would have to be included in the market. Since the penetration rate of mobile broadband was still very low in 2006, we do not investigate whether DSL and cable taken together would be constrained by mobile broadband. As the penetration rates of mobile broadband are increasing rapidly, we think that it is likely to become more relevant in future investigations.

8 Discussion

We use several nested logit discrete choice models to estimate the price elasticity of demand for Internet services in Austria. Our results indicate that demand for broadband Internet access services is rather elastic ($|\varepsilon| > 2.5$ for DSL, cable and mobile) in those areas where several types of broadband access (DSL, cable and mobile) are available. This would indicate that different broadband access technologies are close substitutes and constrain each other. DSL and cable probably form a single market at the retail as well as at the wholesale level. The elasticity of narrowband is lower (−1.93) which may indicate that those users which are still using narrowband do not perceive broadband as an equally good substitute. In areas where only DSL and narrowband are available the DSL elasticity is much lower (−0.97). This suggests that the constraint from narrowband on DSL is limited.

Our findings are consistent with evidence from the US (Crandall et al. 2002; Rappoport et al. 2003) and Portugal (Pereira and Ribeiro 2006) where the demand for DSL also is elastic with a significant cross price elasticity between cable and DSL. Estimates for Japan (with a very high share of DSL-users, see Ida and Kuroda 2006) show, however, that DSL may also form a separate market under particular circumstances. Therefore, a detailed analysis of consumer preferences is necessary before concluding on the appropriate market definition. In any case, national regulatory authorities should closely examine the impact of cable based access (where such access is available) on competition on the wholesale broadband access market either at the level of market definition or at least at the level of market analysis. The impact of other platforms such as FTTH or mobile broadband via UMTS/HSDPA will also warrant attention in some countries.

There are some limitations to the model. One is that the model is static, and therefore switching costs are not allowed for. While installation and set-up fees appear less relevant (many broadband connections are bought in course of regular promotions where installation is free), transaction costs may still be significant. For future estimations it might also be useful to consider demand for service bundles (e.g. broadband with voice and/or TV) since such products are likely to gain importance. A more detailed analysis with regard to fixed-mobile broadband substitution may also become possible in the future since the mobile broadband penetration has increased rapidly in 2007 in Austria (see Cardona et al. 2008).

In most countries (and also in Austria), platforms like cable, FTTH and sometimes also UMTS/HSDPA networks are not available throughout the territory (like DSL networks) but only in cities and densely populated areas. Local loop unbundling activities of alternative operators usually also focus on densely populated areas. This might result in a situation where there is significant competitive pressure on the incumbent DSL operator in some parts of the country while in other parts there is a "DSL monopoly" (only exposed to the limited competitive constraint from narrowband access). The reaction of the incumbent operator to this might be a geographic differentiation of prices in order to be able to compete with other platforms in densely populated areas and at the same time charge the "monopoly price" in the rest of the territory. Most incumbent operators however continue to charge a national uniform price which takes account of the trade-off between the "monopoly price" and the competitive price. In Austria as well as in other countries (see, e.g. Ofcom 2007) this lead to a situation where the incumbent has a quite low retail market share in areas where he competes with other platforms while he holds a very high share in the rest of the country.[26]

Therefore, competitive conditions from a consumer's perspective (price, choice) may differ significantly between areas where other platforms are available and areas where they are not, even if the incumbent operator is setting a national uniform price (and even more if he does price differentiate—see also Cave et al. 2006, p. 29). Consequently there might be a need for a geographically differentiated approach to regulation (if necessary) either by defining different geographical market (such as, for

[26] For a model considering a national operator setting a national uniform price and competing with an entrant with less than national coverage see Valletti et al. (2002).

example, in Ofcom 2007) or by geographically differentiating remedies imposed on the dominant operator.

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Appendix A

Table 7 (a) Bottom stage (cable, DSL, mobile), (b) second stage (narrowband, broadband), (c) third stage (no internet, internet) of the nested logit model for area 1

(a) Bottom stage	
Price	−0.0889 (0.01)***
Download rate	0.0256 (0.01)**
Download volume	0.0042 (0.00)*
Dummy variable for flat tariffs	−3.1182 (0.94)***
Dummy variable for DSL	−1.3232 (0.59)*
Dummy variable for mobile	−1.237 (1.05)
Age interacted with dummy variable for DSL	0.0157 (0.01)
Age interacted with dummy variable for mobile	−0.017 (0.02)
Household size interacted with dummy variable for DSL	0.0584 (0.08)
Household size interacted with dummy variable for mobile	−0.1587 (0.15)
Dummy variable for compulsory school interacted with dummy variable for DSL	0.1682 (0.29)
Dummy variable for high school without graduation interacted with dummy variable for DSL	0.687 (0.32)*
Dummy variable for high school with graduation interacted with dummy variable for DSL	0.2523 (0.29)
Dummy variable for compulsory school interacted with dummy variable for mobile	0.3322 (0.58)
Dummy variable for high school without graduation interacted with dummy variable for mobile	0.9787 (0.6)
Dummy variable for high school with graduation interacted with dummy variable for mobile	0.2862 (0.56)
Dummy variable for female interacted with dummy variable for DSL	0.1169 (0.21)
Dummy variable for female interacted with dummy variable for mobile	−0.0156 (0.4)
Dummy variable for Vienna interacted with dummy variable for DSL	−0.8897 (0.23)***
Dummy variable for Vienna interacted with dummy variable for mobile	−1.1604 (0.45)**
Pseudo R^2	0.338
Number of observations	650
(b) Second stage	
Inclusive value	1.5726 (0.16)***
Dummy variable for narrowband	2.8507 (0.63)***
Age interacted with dummy variable for narrowband	−0.0029 (0.01)
Household size interacted with dummy variable for narrowband	0.0444 (0.06)
Dummy variable for compulsory school interacted with dummy variable for narrowband	−0.5448 (0.25)*
Dummy variable for high school without graduation interacted with dummy variable for narrowband	−0.499 (0.27)
Dummy variable for high school with graduation interacted with dummy variable for narrowband	−0.1955 (0.24)

Table 7 continued

Dummy variable for female interacted with dummy variable for narrowband	−0.1437 (0.17)
Dummy variable for Vienna interacted with dummy variable narrowband	0.0152 (0.21)
Pseudo R^2	0.246
Number of observations	789
(c) Third stage	
Inclusive value	13.4602 (0.95)***
Dummy variable for no Internet	−21.9235 (1.38)***
Age interacted with dummy variable for no Internet	0.0278 (0.00)***
Household size interacted with dummy variable for no Internet	−0.5610 (0.06)***
Dummy variable for compulsory school interacted with dummy variable for no Internet	3.7386 (0.28)***
Dummy variable for high school without graduation interacted with dummy variable for no Internet	1.6175 (0.26)***
Dummy variable for high school with graduation interacted with dummy variable for no Internet	1.2765 (0.26)***
Dummy variable for female interacted with dummy variable for no Internet	0.5408 (0.14)***
Dummy variable for Vienna interacted with dummy variable no Internet	1.7959 (0.19)***
Pseudo R^2	0.406
Number of observations	1830

Standard errors are shown in parentheses. ***, ** and * denote significance on 1%, 5% and 10% level

Table 8 (a) Bottom stage (DSL, narrowband), (b) second stage (no internet, internet) of the nested logit model for area 2

(a) Bottom stage	
Price	−0.0324 (0.01)***
Download rate	0.0769 (0.0113)***
Download volume	0.0171 (0.01)**
Dummy variable for flat rate tariffs	−0.0048 (47.08)
Dummy variable for narrowband	0.1671 (0.21)
Age interacted with dummy variable for narrowband	0.0148 (0.00)***
Household size interacted with dummy variable for narrowband	0.0865 (0.04)**
Dummy variable for compulsory school interacted with dummy variable for narrowband	−0.45 (0.26)
Dummy variable for high school without graduation interacted with dummy variable for narrowband	0.4176 (0.34)
Dummy variable for high school with graduation interacted with dummy variable for narrowband	0.7682 (0.28)***
Dummy variable for female interacted with dummy variable for narrowband	0.1902 (0.12)
Pseudo R^2	0.131
Number of observations	708
(b) Second stage	
Inclusive value	1.9134 (0.55)***
Dummy variable for no Internet	−2.5755 (0.62)***
Age interacted with dummy variable for no Internet	0.0235 (0.01)***
Household size interacted with dummy variable for no Internet	−0.3019 (0.07)***
Dummy variable for compulsory school interacted with dummy variable for no Internet	2.7415 (0.45)***
Dummy variable for high school without graduation interacted with dummy variable for no Internet	1.4814 (0.49)***

Table 8 continued

Dummy variable for high school with graduation interacted with dummy variable for no Internet	−0.1481 (−0.57)
Dummy variable for female interacted with dummy variable for no Internet	0.1577 (0.18)
Pseudo R^2	0.370
Number of observations	532

Standard errors are shown in parentheses. ***, ** and * denote significance on 1%, 5% and 10% level

Table 9 Elasticities

Elasticities	Own price elasticity	Cross price elasticity
Area 1 (cable, DSL, mobile, narrowband)		
Cable	−2.617	0.230
DSL	−2.545	0.402
Mobile	−2.481	0.183
Narrowband	−1.679	0.231
Area 2 (DSL, narrowband)		
DSL	−0.969	0.455
Narrowband	−0.773	0.507

Appendix B: three layer nested logit model

B.1 Choice probabilities

In a nested logit model with three layers, the probability P_{ijlm} that consumer i purchases product j is equal to

$$P_{ijlm} = P_{im} P_{il|m} P_{ij|lm}$$

where we index the first layer alternatives as m and m', the second layer alternatives as l and l', and the third layer alternatives as j and j'. The probability $P_{ij|lm}$ is equal to

$$P_{ij|lm} = \frac{e^{V_{ijlm}/\lambda_{lm}}}{\sum_{j'} e^{V_{ij'lm}/\lambda_{lm}}}.$$

The probability $P_{il|m}$ is equal to

$$P_{il|m} = \frac{e^{V_{lm}/\lambda_m + \lambda_{lm} IV_{lm}/\lambda_m}}{\sum_{l'} e^{V_{l'm}/\lambda_m + \lambda_{l'm} IV_{l'm}/\lambda_m}}$$

with the inclusive value IV_{lm} to be equal

$$IV_{lm} = \ln \sum_{j'} e^{V_{j'lm}/\lambda_{lm}}.$$

The probability P_{im} is equal to

$$P_{im} = \frac{e^{V_m + \lambda_m IV_m}}{\sum_{m'} e^{V_{m'} + \lambda_{m'} IV_{m'}}}$$

with the inclusive value IV(m) equal to

$$IV_m = \ln \sum_{l'} e^{V_{l'm}/\lambda_m + \lambda_{l'm} IV_{l'm}/\lambda_m}.$$

B.2 Derivation of elasticities

Let us define the own price elasticity to be $\mu(jj)$. It is equal to

$$\mu_{jj} = \frac{\partial P_{jlm}}{\partial X_{jlm}} \frac{X_{jlm}}{P_{jlm}},$$

with P_{jlm} the probability of choice j and X_{jlm} the price of that choice. We index the first layer alternatives as m and m', the second layer alternatives as l and l', and the third layer alternatives as j and j'. For simplicity, we suppress the consumer-specific index i.

The probability P_{jlm} is equal to

$$P_{jlm} = P_m P_{l|m} P_{j|lm}.$$

Its derivative with respect to X_{jlm} is equal to

$$\frac{\partial P_{jlm}}{\partial X_{jlm}} = \frac{\partial P_m}{\partial X_{jlm}} P_{l|m} P_{j|lm} + P_m \frac{\partial P_{l|m}}{\partial X_{jlm}} P_{j|lm} + P_m P_{l|m} \frac{\partial P_{j|lm}}{\partial X_{jlm}}$$

The probability $P_{j|lm}$ is equal to

$$P_{j|lm} = \frac{e^{V_{j|m}/\lambda_{lm}}}{\sum_{j'} e^{V_{j'lm}/\lambda_{lm}}}.$$

The probability $P_{l|m}$ is equal to

$$P_{l|m} = \frac{e^{V_{lm}/\lambda_m + \lambda_{lm} IV_{lm}/\lambda_m}}{\sum_{l'} e^{V_{l'm}/\lambda_m + \lambda_{l'm} IV_{l'm}/\lambda_m}}$$

with the inclusive value

$$IV_{lm} = \ln \sum_{j'} e^{V_{j'lm}/\lambda_{lm}}.$$

The probability P_m is equal to

$$P_m = \frac{e^{V_m + \lambda_m IV_m}}{\sum_{m'} e^{V_{m'} + \lambda_{m'} IV_{m'}}}$$

with the inclusive value

$$IV_m = \ln \sum_{l'} e^{V_{l'm}/\lambda_m + \lambda_{l'm} IV_{l'm}/\lambda_m}.$$

The derivative of $P_{j|lm}$ with respect to X_{jlm} is equal to

$$\frac{\partial P_{j|lm}}{\partial X_{jlm}} = \frac{\partial V_{j|lm}}{\partial X_{jlm}} \frac{1}{\lambda_{lm}} P_{j|lm}(1 - P_{j|lm})$$

The derivative of $P_{l|m}$ with respect to X_{jlm} is equal to

$$\frac{\partial P_{l|m}}{\partial X_{jlm}} = \frac{\partial V_{j|lm}}{\partial X_{jlm}} \frac{1}{\lambda_m} P_{l|m}(1 - P_{l|m}) P_{j|lm}$$

The derivative of P_m with respect to X_{jlm} is equal to

$$\frac{\partial P_m}{\partial X_{jlm}} = \frac{\partial V_{j|lm}}{\partial X_{jlm}} P_m(1 - P_m) P_{l|m} P_{j|lm}$$

The own price elasticity is then equal to

$$\mu_{jj} = \frac{\partial V_{j|lm}}{\partial X_{jlm}} \frac{1}{\lambda_{lm}} X_{jlm} \left[(1 - P_m) P_{l|m} \lambda_{lm} P_{j|lm} + (1 - P_{l|m}) \frac{\lambda_{lm}}{\lambda_m} P_{j|lm} + (1 - P_{j|lm}) \right]$$

Let us define the cross price elasticity to be $\mu(j'j)$. It is equal to

$$\mu_{j'j} = \frac{\partial P_{j'lm}}{\partial X_{jlm}} \frac{X_{jlm}}{P_{j'lm}}$$

with $P_{j'lm}$ the probability of choice j' and X_{jlm} the price of an alternative choice.

The probability $P_{j'lm}$ is equal to

$$P_{j'lm} = P_m P_{l|m} P_{j'|lm}$$

Its derivative with respect to X_{jlm} is equal to

$$\frac{\partial P_{j'lm}}{\partial X_{jlm}} = \frac{\partial P_m}{\partial X_{jlm}} P_{l|m} P_{j'|lm} + P_m \frac{\partial P_{l|m}}{\partial X_{jlm}} P_{j'|lm} + P_m P_{l|m} \frac{\partial P_{j'|lm}}{\partial X_{jlm}}$$

The derivative of $P_{j'|lm}$ with respect to X_{jlm} is equal to

$$\frac{\partial P_{j'|lm}}{\partial X_{jlm}} = \frac{\partial V_{j|lm}}{\partial X_{jlm}} P_{j|lm} P_{j'|lm}$$

The cross price elasticity is then equal to

$$\mu_{j'j} = -\frac{\partial V_{j|lm}}{\partial X_{jlm}} \frac{1}{\lambda_{lm}} X_{jlm} \left[(1 - P_m)\lambda_{lm} P_{l|m} P_{j|lm} + (1 - P_{l|m}) \frac{\lambda_{lm}}{\lambda_m} P_{j|lm} + P_{j|lm} \right]$$

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